Year 11 · AQA Level 2 Further Maths · Lesson 3

Simultaneous equations: linear-quadratic and three unknowns

w/c 9 November 2026 · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Solve a linear and a quadratic equation together

2. Solve three linear equations with three unknowns

Do now: retrieval practice

4 mins

1 Solve $x + y = 7$ and $x - y = 1$

$x = 4, y = 3$

2 Expand $(x + 2)^2$

$x^2 + 4x + 4$

3 Factorise $x^2 + x - 6$

$(x + 3)(x - 2)$

4 Make y the subject of $2x + y = 5$

$y = 5 - 2x$

Objective 1

Solve a linear and a quadratic equation together

Key idea

Learn

!

Rearrange the linear equation to make one letter the subject, then substitute it into the quadratic. Solve the quadratic, then put each answer back into the LINEAR equation to find the pairs.

Objective 1: Solve a linear and a quadratic equation together

I do: example 1

I do

Solve $y = x^2 - 3$ and $y = 2x$

1. $x^2 - 3 = 2x$
2. $x^2 - 2x - 3 = 0$, so $(x - 3)(x + 1) = 0$
3. $x = 3$ gives $y = 6$; $x = -1$ gives $y = -2$
4. $(3, 6)$ and $(-1, -2)$

I do: example 2

I do

Solve $x^2 + y^2 = 25$ and $y = x + 1$

1. $x^2 + (x + 1)^2 = 25$
2. $2x^2 + 2x - 24 = 0$, so $x^2 + x - 12 = 0$
3. $(x + 4)(x - 3) = 0$
4. $x = 3, y = 4$ or $x = -4, y = -3$

I do: example 3

I do

Solve $xy = 6$ and $x + y = 5$

1. $y = 5 - x$
2. $x(5 - x) = 6$, so $x^2 - 5x + 6 = 0$
3. $x = 2$ or $x = 3$
4. (2, 3) and (3, 2)

Mini whiteboards

We do

1. Solve $y = x^2$ and $y = x + 2$

Answer: (2, 4) and (−1, 1)

2. Solve $x^2 + y^2 = 10$ and $y = x + 2$

Answer: (1, 3) and (−3, −1)

3. Solve $y = x^2 - 1$ and $y = 3$

Answer: (2, 3) and (−2, 3)

4. Solve $xy = 8$ and $y = x + 2$

Answer: (2, 4) and (−4, −2)

You do: choose your challenge

You do

Bronze

Solve $y = x^2$ and $y = 4x - 3$

Silver

Solve $y = x^2 + 2x$ and $y = x + 6$

Gold

Solve $x^2 + y^2 = 20$ and $y = 2x$

Platinum

Solve $x^2 + y^2 = 34$ and $x + y = 8$

You do: answers

Check

Bronze (1, 1) and (3, 9)

Silver (2, 8) and (−3, 3)

Gold (2, 4) and (−2, −4)

Platinum (3, 5) and (5, 3)

If you finish early

Extra

Bronze

Solve $y = x^2$ and $y = 9$

(3, 9) and (−3, 9)

Silver

Solve $y = x^2 - 2x$ and $y = x + 4$

(4, 8) and (−1, 3)

Gold

Solve $x^2 + y^2 = 50$ and $y = x$

(5, 5) and (−5, −5)

Platinum

Solve $x^2 + y^2 = 26$ and $x - y = 4$

(5, 1) and (−1, −5)

Exam question 1

4 marks

Solve $y = x^2 - 4x + 5$ and $y = 2x - 3$

Space for working

Mark scheme: $x^2 - 6x + 8 = 0$ (M1). $(x - 2)(x - 4) = 0$ (M1). $x = 2$, $y = 1$ (A1). $x = 4$, $y = 5$ (A1)

Exam question 2

4 marks

The line $y = 3 - x$ meets the circle $x^2 + y^2 = 9$ at A and B. Find A and B.

Space for working

Mark scheme: $x^2 + (3 - x)^2 = 9$ (M1). $2x^2 - 6x = 0$ (M1). $x = 0$ or $x = 3$ (A1). A(0, 3) and B(3, 0) (A1)

Objective 1 achieved

Solve a linear and a quadratic equation together



Objective 2

Solve three linear equations with three unknowns

✓ Achieved: Solve a linear and a quadratic equation together

Key idea

Learn

!

Number the equations. Use two different pairs to eliminate the SAME letter, giving two equations in two unknowns. Solve those, then substitute back to find the third letter. Check in all three.

Objective 2: Solve three linear equations with three unknowns

I do: example 1

I do

Solve $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$

1. (1) − (2): $2y = 4$, so $y = 2$
2. (1) + (3): $3x + 2y = 7$, so $3x = 3$ and $x = 1$
3. Substitute into (1): $1 + 2 + z = 6$, so $z = 3$
4. $x = 1$, $y = 2$, $z = 3$

I do: example 2

I do

Solve $2x + y - z = 2$, $x + 2y + z = 7$, $3x - y + 2z = 11$

1. $(1) + (2)$: $3x + 3y = 9$, so $x + y = 3$
2. $2 \times (1) + (3)$: $7x + y = 15$
3. Subtract: $6x = 12$, so $x = 2$ and $y = 1$
4. In (2): $2 + 2 + z = 7$, so $z = 3$

I do: example 3

I do

Three numbers add to 15. The first is twice the second. The third is 3 more than the second. Find them.

1. $a + b + c = 15$, $a = 2b$, $c = b + 3$
2. $2b + b + (b + 3) = 15$
3. $4b = 12$, so $b = 3$
4. $a = 6$, $b = 3$, $c = 6$

Mini whiteboards

We do

1. Solve $x + y + z = 9$, $y + z = 7$, $z = 4$

Answer: $x = 2, y = 3, z = 4$

2. Solve $x + y + z = 11$, $x - y = 1$, $y - z = 2$

Answer: $x = 5, y = 4, z = 2$

3. Solve $x + y + z = 12$, $y + z = 8$, $z = 3$

Answer: $x = 4, y = 5, z = 3$

4. Solve $x + y + z = 9$, $x + y = 5$, $x - y = 1$

Answer: $x = 3, y = 2, z = 4$

You do: choose your challenge

You do

Bronze

Solve $x + y + z = 6$, $y + z = 5$, $z = 3$

Silver

Solve $x + y + z = 10$, $x - y = 2$, $z = 2x$

Gold

Solve $x + y + z = 6$, $2x + y - z = 1$, $x - y + 2z = 5$

Platinum

Solve $a + b + c = 0$, $a + 2b + 4c = 3$, $a + 3b + 9c = 8$

You do: answers

Check

Bronze

x = 1, y = 2, z = 3

Silver

x = 3, y = 1, z = 6

Gold

x = 1, y = 2, z = 3

Platinum

a = −1, b = 0, c = 1

If you finish early

Extra

Bronze

Solve $x + y + z = 15$, $y + z = 10$, $z = 6$

$x = 5, y = 4, z = 6$

Silver

Solve $x + y + z = 7$, $x - y = 1$, $x + z = 5$

$x = 3, y = 2, z = 2$

Gold

Solve $x + y + z = 2$, $2x - y + z = 7$, $x + y - z = -4$

$x = 1, y = -2, z = 3$

Platinum

1 tea + 1 coffee + 1 cake = £7. 2 teas + 1 coffee = £6.50. 1 coffee + 2 cakes = £7.50. Find each price.

Tea £1.50, coffee £3.50, cake £2

Exam question 1

4 marks

Solve $x + y + z = 4$, $2x - y + z = 8$, $x + 2y - z = -3$

Space for working

Mark scheme: (1) + (3): 2x + 3y = 1 (M1). (2) + (3): 3x + y = 5 (M1). x = 2, y = −1 (A1). z = 3 (A1)

Exam question 2

3 marks

Solve $y = x^2 + 1$ and $y = 3x - 1$

Space for working

Mark scheme: $x^2 - 3x + 2 = 0$ (M1). $x = 1$ or $x = 2$ (A1). (1, 2) and (2, 5) (A1)

Objective 2 achieved

Solve three linear equations with three unknowns



Real life

Apply

Tickets: 2 adult + 3 child + 1 senior cost £58. 1 adult + 1 child + 2 senior cost £44. 3 adult + 1 child + 1 senior cost £60. Find each price.

Adult £14, child £6, senior £12

Extension: stretch yourself

Stretch

1. Find k so that the line $y = 2x + k$ touches the curve $y = x^2 + 3$ exactly once.

$x^2 - 2x + 3 - k = 0$ has one root: $4 - 4(3 - k) = 0$, so $k = 2$, touching at $(1, 4)$

2. Solve $x^2 + y^2 = 13$ and $xy = 6$

$(2, 3)$, $(3, 2)$, $(-2, -3)$ and $(-3, -2)$

Exit ticket: exam questions

Exit

1. Solve $y = x^2 - 2$ and $y = x$ **(2 marks)**

2. Solve $x^2 + y^2 = 5$ and $y = x + 1$ **(2 marks)**

3. Solve $x + y + z = 7$, $x - y = 1$, $y + z = 5$ **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $x^2 - x - 2 = 0$ (M1). (2, 2) and (−1, −1) (A1) [2]

Q2 $2x^2 + 2x - 4 = 0$ (M1). (1, 2) and (−2, −1) (A1) [2]

Q3 x = 2, y = 1, z = 4 (B2) [2]

Total: ____ / 6

Today I achieved

- ✔ **Solve a linear and a quadratic equation together**
- ✔ **Solve three linear equations with three unknowns**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Solve $y = x^2 + 4$ and $y = 5x$ [3]

2. Solve $x^2 + y^2 = 17$ and $y = x + 3$ [4]

3. Solve $xy = 12$ and $x + y = 7$ [3]

4. Solve $x + y + z = 8$, $x + y - z = 2$, $x - y + z = 4$ [3]

5. Solve $x + 2y + z = 9$, $2x + y - z = 0$, $x - y + 2z = 7$ [4]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Simultaneous equations"